TECHNICAL ARTICLE

Quantum-Resistant security in V2X and VANET:

A Structured Approach to PQC Migration

About this article

Current vehicular communication security was designed for a pre-quantum world. This article examines where the cryptographic fault lines are, which use cases demand attention first, how geography shapes the migration strategy, and what realistic outcomes look like when PQC meets the hard real-time constraints of connected vehicles.

01 The problem with today's cryptographic baseline

Vehicular networks were architected under a very specific and now-expiring assumption: that the mathematics underpinning public-key cryptography, specifically the difficulty of factoring large integers and solving discrete logarithm problems, would remain computationally intractable for any foreseeable adversary. That assumption breaks the moment a cryptographically relevant quantum computer (CRQC) becomes operational.

Unlike most industries, the automotive and ITS sector faces a compounding version of this risk. A vehicle manufactured today has a service life of 12–20 years. Firmware and security credential infrastructure baked into OBUs today will still be operating when CRQC availability is projected to move from speculative to probable, conservatively estimated between 2030 and 2035 by multiple national security agencies.

HARVEST NOW, DECRYPT LATER

Adversaries are already capturing encrypted V2X traffic with the intent to decrypt it retroactively once quantum capabilities become available. For data that has long-term sensitivity, identity credentials, location history, law enforcement triggers, and military logistics, the threat is not future-tense. It is present-tense.

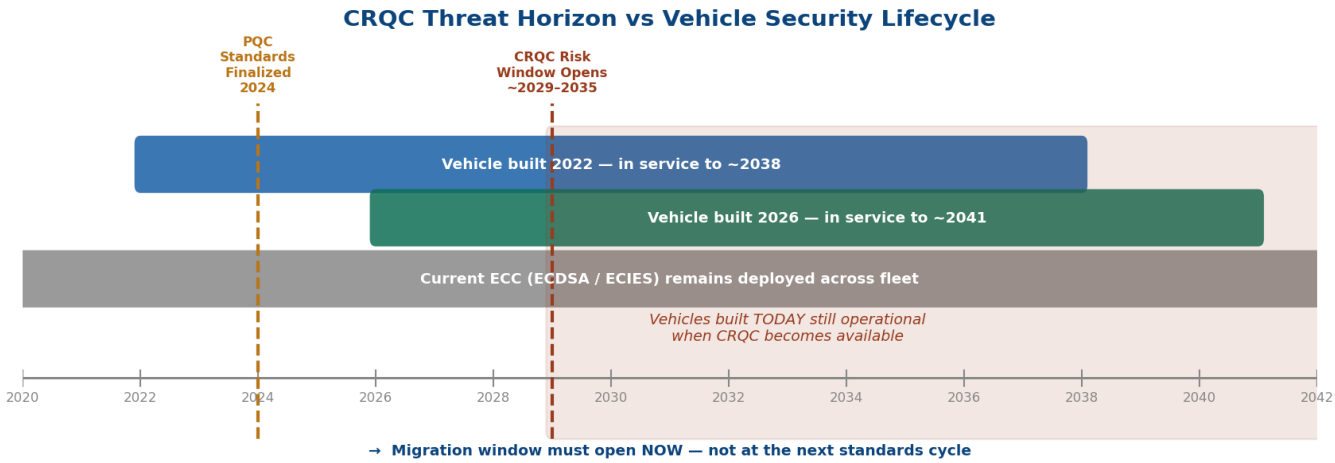

Figure 1.1 CRQC threat horizon vs vehicle security lifecycle overlap (2024–2040). Vehicles manufactured today remain in service when quantum risk becomes acute.

This is not a speculative risk that can be deferred to a future standards cycle. The long hardware lifecycle of vehicles, the difficulty of over-the-air updates to HSMs, and the sheer scale of PKI

infrastructure underpinning SCMS and C-ITS deployments mean the migration window must open now.

02 Current encryption standards in V2X and VANET

The cryptographic foundation of V2X is remarkably consistent across geographic standards bodies. Both primary frameworks, the North American WAVE/DSRC stack and the European ETSI C-ITS stack, anchor on the same fundamental primitive: elliptic curve cryptography (ECC).

Table 1: Current V2X security standards and quantum vulnerability status

Standard / Framework	Region	Signature scheme	Key exchange	Sym. Enc.	Quantum status
IEEE 1609.2 / WAVE	North America	ECDSA (NIST P-256)	ECIES	AES-CCM 128	**Vulnerable**
ETSI TS 103 097 / C-ITS	Europe	ECDSA (P-256/Brainpool)	ECIES	AES-CCM	**Vulnerable**
SAE J2945/1	North America	ECDSA P-256 (BSMs)	Per IEEE 1609.2	AES-CCM	**Vulnerable**
3GPP C-V2X (PC5/Uu)	Global	ECC (SIM-based)	LTE/5G-AKA	AES/SNOW 128	**Partial risk**
ISO/SAE 21434	Global	Process standard only			**Framework only**
UNECE WP.29 (R155/R156)	Global	Regulatory only			**Framework only**

Key insight

ECDSA on NIST P-256 is directly broken by Shor's algorithm running on a CRQC. AES-128 is weakened but not broken by Grover's algorithm; moving to AES-256 provides adequate symmetric security. The critical exposure is in authentication and key exchange, not bulk encryption.

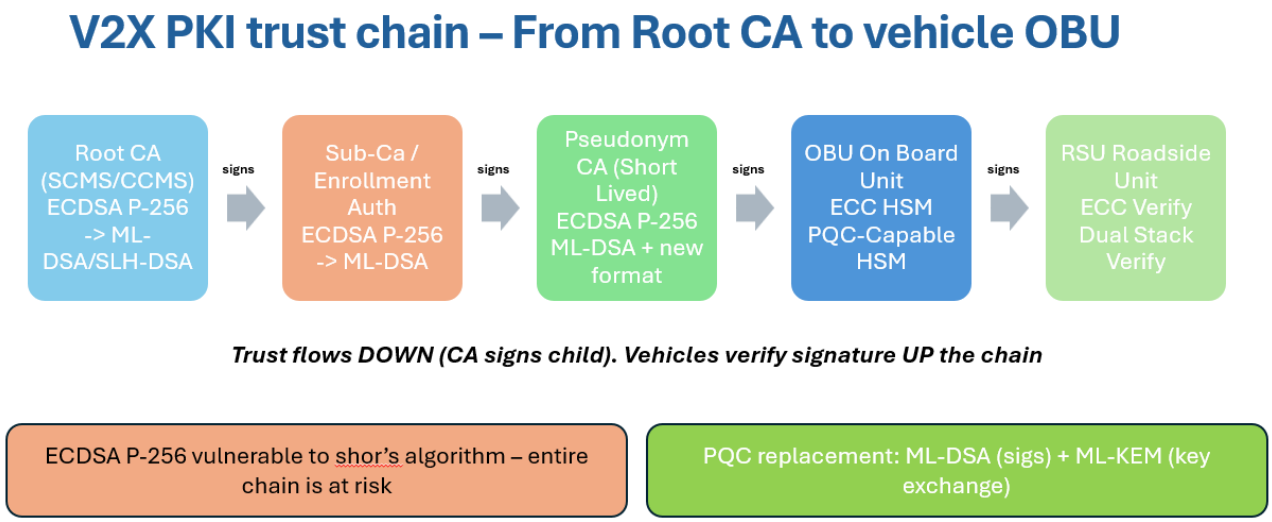

Figure 2.1 V2X PKI trust chain showing the full credential hierarchy from Root CA to OBU, with current algorithms and their PQC replacements at each level.

03 What PQC actually means in this context

NIST finalised its primary post-quantum cryptographic standards in August 2024, with a fourth algorithm (HQC) selected as a backup KEM in March 2025. The table below summarises the primary algorithms and their relevance to V2X.

Table 2: NIST PQC algorithms and V2X applicability

Algorithm	FIPS	Type	Replaces	V2X Fit
ML-KEM (Kyber)	203	KEM lattice (Module-LWE)	ECDH / ECIES	Good compact keys, fast. Session key establishment (V2I, V2C).
ML-DSA (Dilithium)	204	Signature lattice (Module-LWE)	ECDSA	Medium signature size 2,420B. Best for CA/OTA signing, not 10 Hz beaconing.
SLH-DSA (SPHINCS+)	205	Signature hash-based, stateless	ECDSA (CA only)	CA layer only 8-50 KB signatures, too large for V2V messages.
FN-DSA (FALCON)	206*	Signature lattice (NTRU)	ECDSA	Best V2V fit smallest PQC signature (~690B). Complex state management.

The v2x-specific tension

PQC algorithms are not drop-in replacements. ML-DSA signatures are 2–7x larger than those of ECDSA. The IEEE 1609.2 certificate format was explicitly designed to be compact, smaller than X.509, because V2X packets are small and high-frequency. Every byte added to a safety message header has a measurable effect on latency and channel congestion at scale.

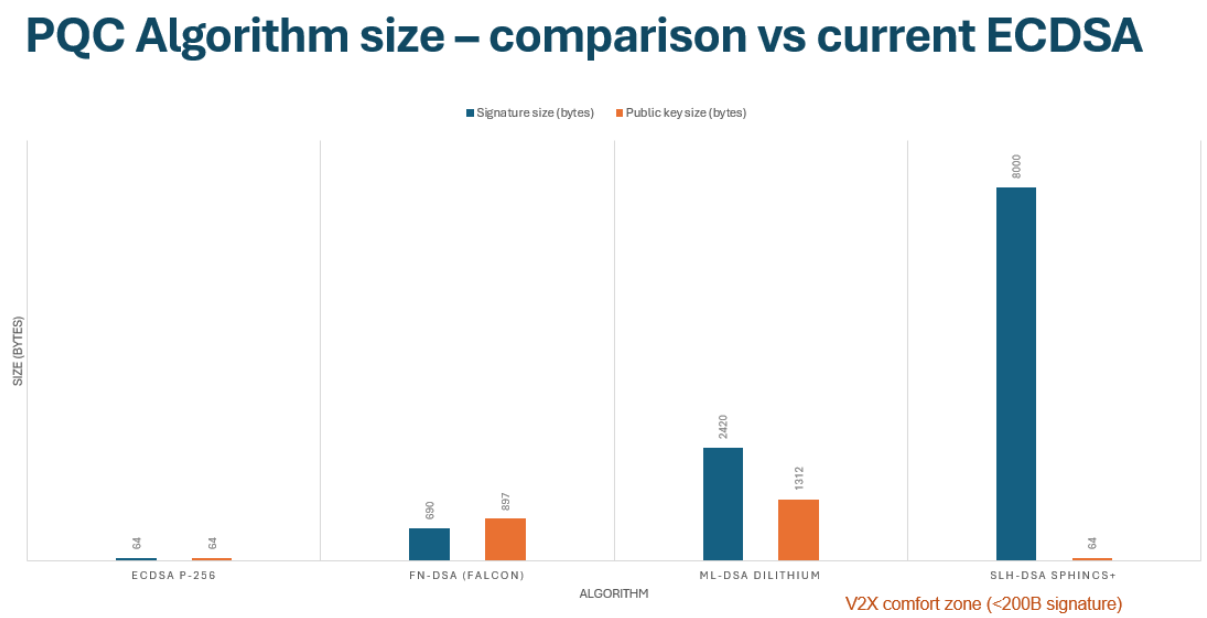

Figure 3.1 Signature and public key size comparison: current ECDSA vs NIST PQC algorithms. The V2X comfort zone (< 200 bytes) is indicated.

04 Use Cases and where to prioritise

Not all V2X use cases have the same quantum risk profile. Migration priority should be determined by data sensitivity, credential longevity, and consequence of compromise.

Table 3: V2X Use Case PQC priority classification

Use Case	Message Type	Sensitivity	Freq.	PQC Priority
PKI credential provisioning (SCMS/CCMS)	Certificate issuance, CRL	Very high root of trust	Low	**Critical first**
OTA firmware/software updates	Signed packages (V2C)	Very high code integrity	Low	**Critical first**
Emergency vehicle preemption (EVP)	Signed SPaT/BSM	High safety-critical	Med	**High**
Traffic signal control (V2I SPaT/MAP)	Authenticated infra msgs	High infrastructure trust	High	**High**
Toll/access control (V2I)	Identity + payment tokens	High financial + privacy	Low	**High**
V2V safety beaconing (BSM/CAM)	Pseudonym-signed broadcast	Medium ephemeral identity	10 Hz	**Medium**
Fleet telematics / remote diagnostics	TLS-protected TCP/IP (V2C)	Medium operational data	Med	**Medium**
V2P pedestrian safety alerts	Signed broadcast	Low-Medium	Med	**Medium**
In-vehicle infotainment/navigation	TLS / HTTPS	Low	Var	**Lower**

The asymmetry principle

Infrastructure-side operations (CA signing, OTA delivery, credential provisioning) happen infrequently, involve powerful hardware, and have a large impact if compromised. These should be migrated to PQC first. The hard problem is high-frequency V2V beaconing, where message-rate and latency constraints directly collide with PQC overhead.

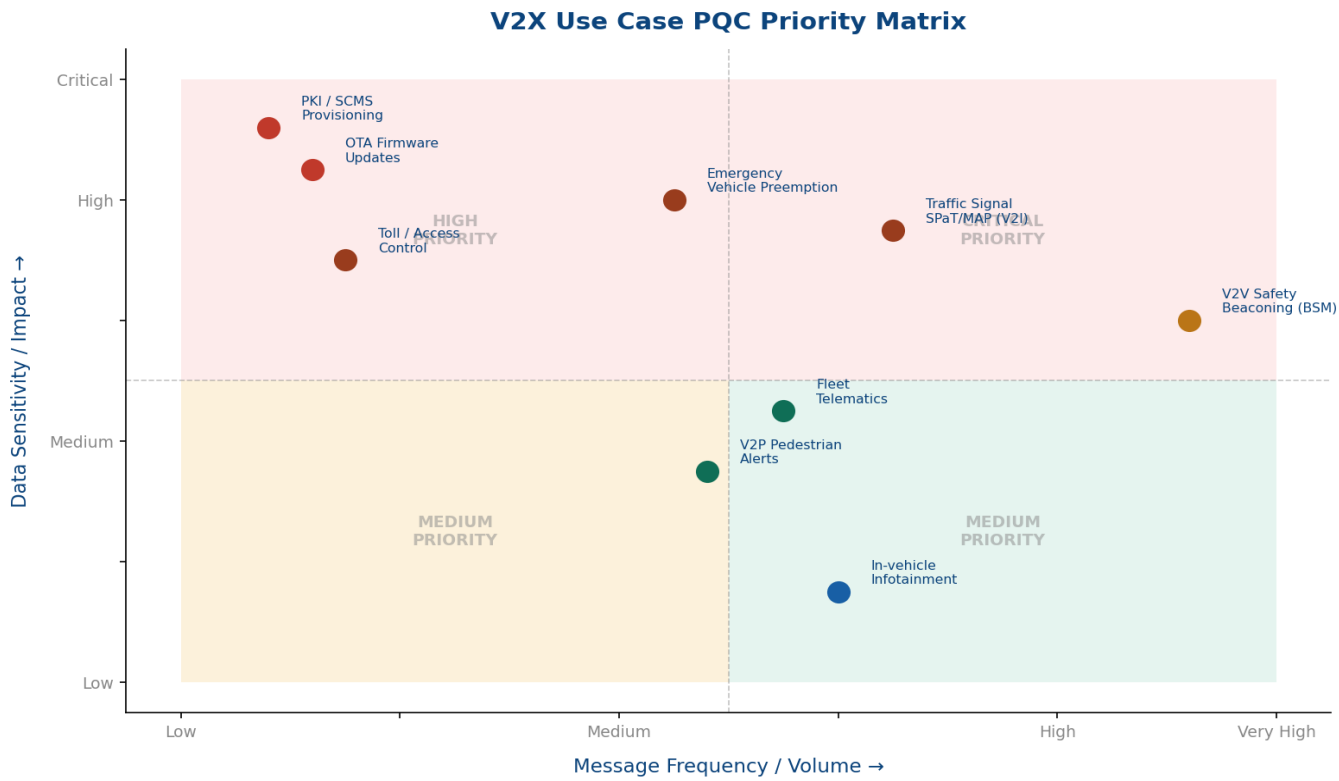

Figure 4.1 V2X use case priority matrix: data sensitivity vs message frequency. Critical-priority use cases (upper-right zone) should lead the migration.

05 Geographic and Regional boundaries

The question of whether PQC deployment in V2X should be geographically scoped has direct implications for interoperability, regulatory compliance, and deployment complexity. The answer is yes, but with caveats about cross-border continuity.

Table 4: Regional PQC landscape for V2X

Region	Standards Body	PQC Timeline Status	Key Considerations
North America	NIST / USDOT / IEEE	FIPS finalised Aug 2024. Federal mandate: transition by 2035 (NSS by 2030).	SCMS is centralised enough for a phased upgrade. Smart corridors are natural pilots.
Europe (EU)	ETSI / EC / CCMS	ETSI TC ITS WG5 is working on PQC for C-ITS. No fixed mandate date yet.	CCMS spans member states' single root CA transition cascades. GDPR complexity.
China / APAC	OSCCA / GB/T standards	China follows SM2/SM3/SM4, which may diverge from NIST selections.	Cross-border (GBA, ASEAN) fragmentation risk. Japan/Korea are more NIST-aligned.
GCC / Middle East	National authorities	Greenfield smart city deployments best opportunity for native PQC.	UAE, Saudi Arabia, Qatar quantum readiness strategies are active.

Cross-border corridors	Bilateral agreements	No unified cross-border PQC V2X framework yet.	Hybrid certs and bilateral SCMS trust agreements are needed for continuity.

Defining practical geographic scope

A defensible approach is to define initial PQC deployment zones by existing ITS deployment footprints: smart corridors, connected intersection deployments, and autonomous vehicle operational design domains (ODDs). These are already bounded, infrastructure-equipped zones with managed RSU populations ideal for a controlled PQC rollout before scaling to national or regional fleets.

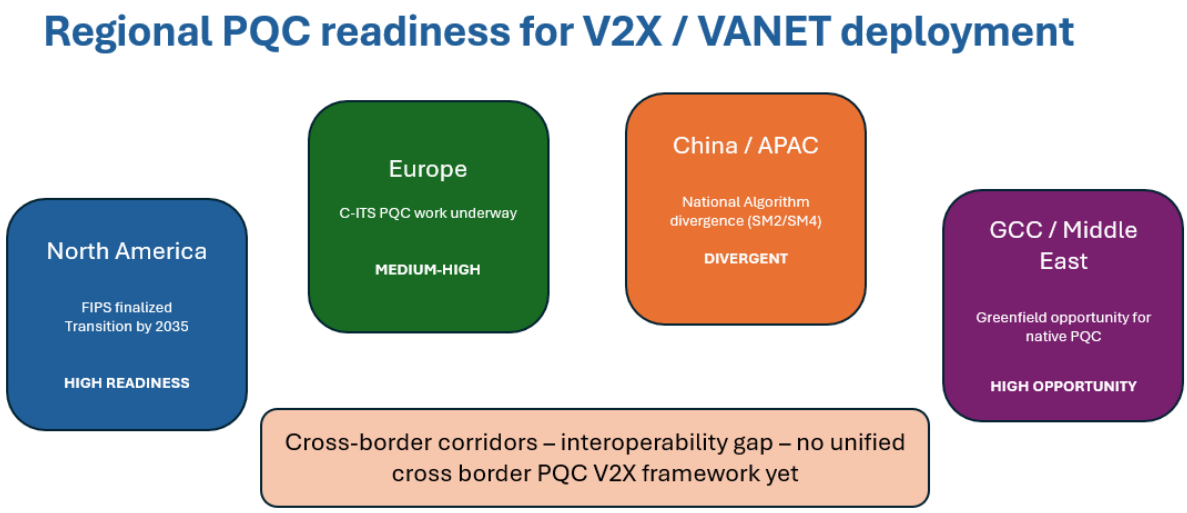

Figure 5.1 Regional PQC readiness for V2X deployment. North America leads with finalised NIST standards. GCC offers a greenfield native-PQC opportunity.

Defining deployment boundaries is not just a regulatory exercise; it directly shapes the SCMS architecture. A geographic boundary creates an implicit trust domain. PQC-capable vehicles need a certificate authority chain that terminates at a PQC root CA recognised by all entities within that boundary.

06 Where PQC must be applied - The layer map

PQC is not a single switch that can be flipped on or off in the V2X stack. It needs to be applied at multiple layers, each with different constraints and migration complexity.

Table 5: V2X Layer Map - Where PQC applies and migration difficulty

Layer	Component	Current algorithm	PQC replacement	Difficulty
PKI / Trust	Root CA, Sub-CA, Registration Authority	ECDSA P-256	ML-DSA or SLH-DSA	**Medium**
PKI / Trust	Pseudonym certificate issuance	ECDSA P-256 (implicit)	ML-DSA + cert format redesign	**High**
Message Auth	V2V BSM / CAM signatures (10 Hz)	ECDSA	FN-DSA or hybrid ECC+PQC	**Very high**

Key Exchange	Session key establishment (V2I, V2C)	ECIES (ECC-based)	ML-KEM	**Medium**
Bulk Encryption	Payload encryption	AES-128-CCM	AES-256-CCM	**Low parameter change**
OTA Updates	Firmware package signing (V2C)	ECDSA / RSA	ML-DSA or SLH-DSA	**Medium**
TLS / HTTPS	Backend V2C communications	TLS 1.3 + ECDHE	TLS 1.3 + ML-KEM hybrid	**Low-Open SSL eco**
HSM / OBU	On-board hardware crypto (silicon)	ECC accelerator (SAF5100)	PQC-capable HSM (new silicon)	**Very high**

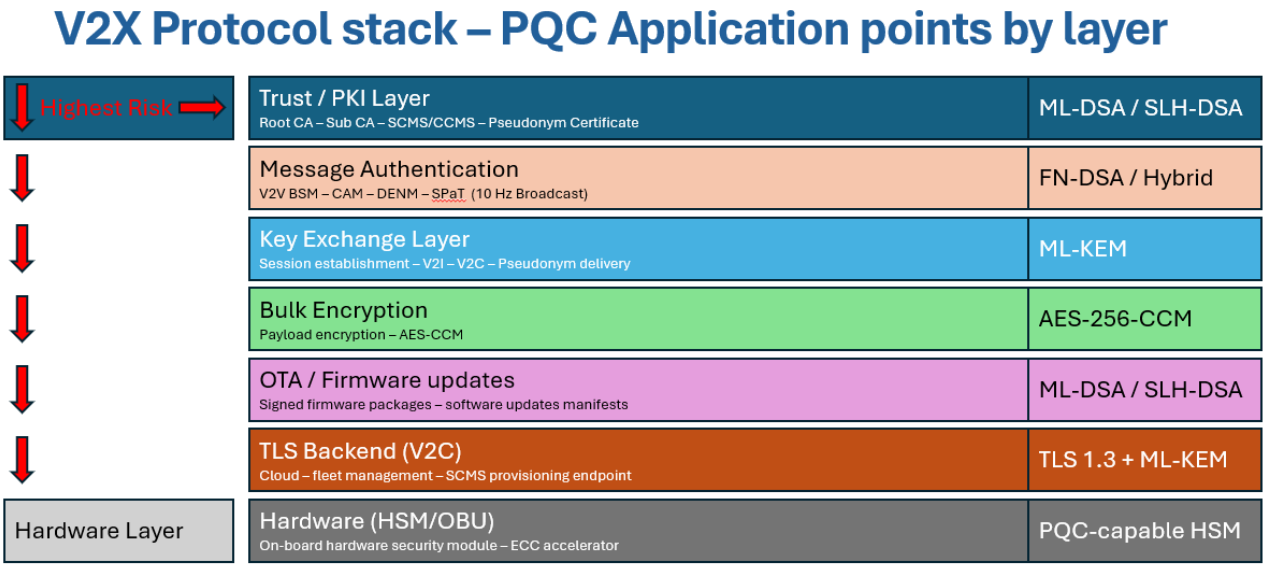

Figure 6.1 V2X protocol stack annotated with PQC application points. Each layer shows the current algorithm, PQC replacement, and relative migration difficulty.

07 Challenges

Seven structural challenges must be confronted for any credible PQC migration in V2X. These represent concrete engineering, governance, and standards problems that do not yet have complete solutions.

C1

Latency vs. security overhead

Safety-critical V2V messages must be verified within 100 ms end-to-end. The current ECDSA on a hardware security module takes 2.5–10 ms. Standard ML-DSA introduces significantly more computation. Research into adaptive PQC selection using reinforcement learning shows promise but adds system complexity. FALCON's smaller signatures (~690 bytes vs. ML-DSA's ~2,420 bytes) are the most viable path for high-frequency safety messages, but its key management introduces failure modes if not implemented carefully.

C2

Certificate and message size inflation

PQC certificates will be substantially larger than the current IEEE 1609.2 implicit certificates. The V2X channel operates at 6–27 Mbps over DSRC/PC5, with high message density in traffic-dense environments. The certificate format in IEEE 1609.2 was explicitly designed to

be smaller than the X.509 format. A PQC migration may require a new format specification, not merely algorithm substitution.

C3

Hardware lifecycle and OBU constraints

Existing deployed OBUs contain dedicated ECC hardware accelerators (e.g., SAF5100 ECDSA verification accelerator). These cannot be software-patched to support PQC. New silicon with PQC acceleration is needed. Vehicle replacement cycles of 10–20 years mean a dual-stack (ECC + PQC hybrid) approach must be maintained for a long transition period.

C4

SCMS / CCMS infrastructure migration

The North American SCMS and European CCMS are the foundational trust anchors for all V2X certificates. Migrating these to PQC root CAs requires coordinated multi-party upgrades across governments, operators, and automakers. A root CA key rollover is a high-stakes operation; if done incorrectly, it can invalidate millions of in-field vehicle certificates simultaneously.

C5

Regulatory and standards fragmentation

NIST, ETSI, 3GPP, SAE, IEEE, OSCCA, and ISO/SAE 21434 are not synchronised on PQC timelines. A vehicle certified to comply with IEEE 1609.2-2022 cannot simply be updated to ML-DSA without triggering re-certification. UN R155 requires cybersecurity management systems to be maintained in a demonstrably secure manner. A significant PQC upgrade constitutes a cybersecurity-relevant change requiring OEM documentation.

C6

Privacy preservation under PQC

The pseudonym certificate model is designed to prevent vehicle tracking while enabling authentication. Larger PQC certificates reduce the number of pseudonyms that can be economically provisioned and stored in an OBU. Certificate rotation strategies that protect privacy must be redesigned for a larger-certificate world. GDPR adds a legal dimension to any change in certificate handling.

C7

Algorithm maturity and side-channel risk

NIST's PQC standards are new. ECDSA has decades of implementation experience and known side-channel mitigations. Lattice-based algorithms have less implementation history in constrained embedded environments. Timing, side-channel, power analysis, and fault-injection attacks against PQC in automotive-grade HSMs are active research areas. Premature deployment without implementation hardening creates new attack surfaces.

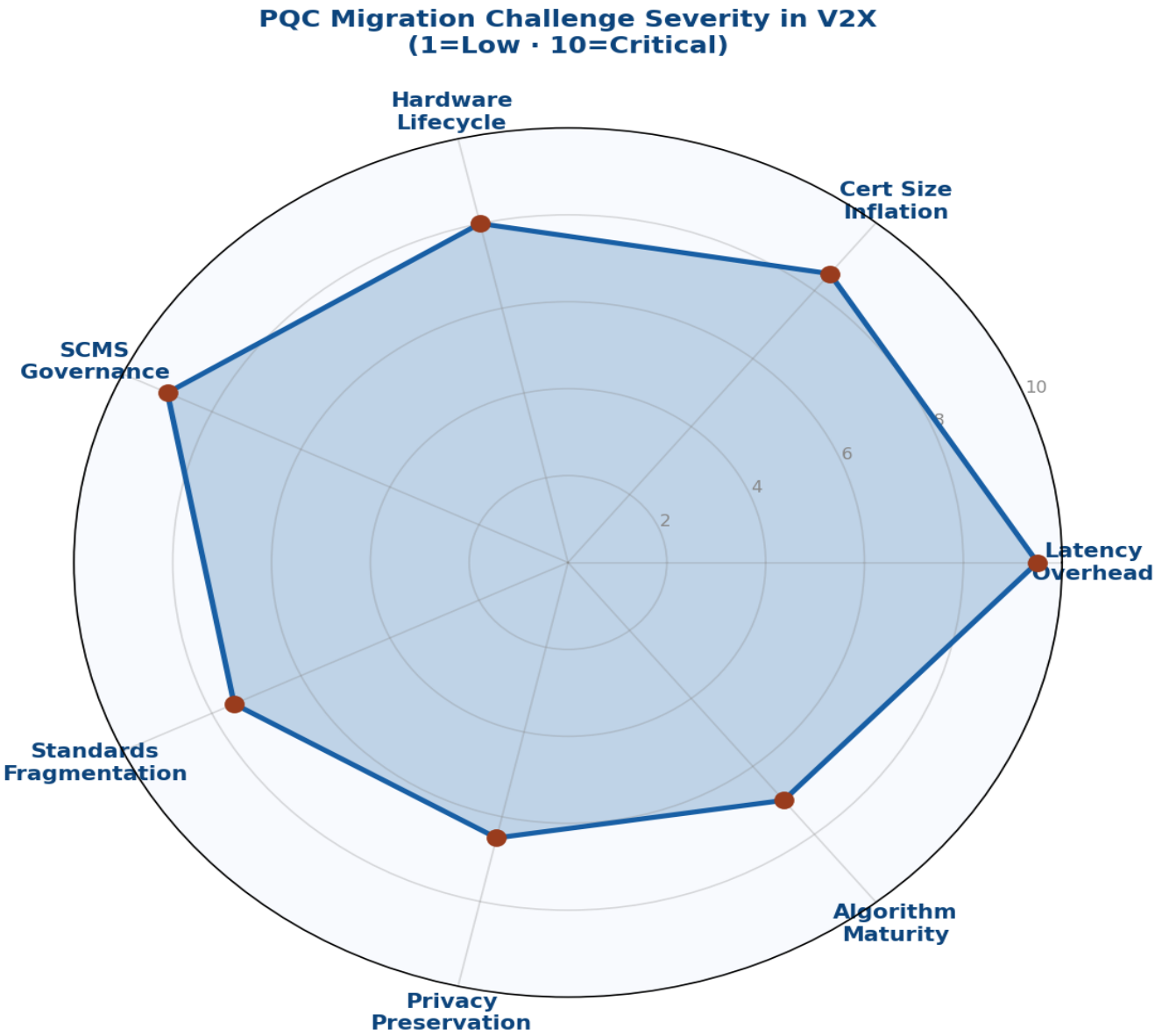

Figure 7.1 Radar chart: Seven challenge dimensions for PQC migration in V2X, rated by severity (1=Low, 10=Critical). Latency and governance rank highest.

08 Expected outcomes vs. What is realistic

The most intellectually honest position is that full PQC migration of V2X, particularly the high-frequency V2V safety beaconing layer, will not be completed before 2035 across any major region's deployed fleet.

Table 6: Literature promises vs. Near-Term reality (2026–2030)

What the literature promises	What is currently realistic (2026–2030)
Quantum-resistant authentication for all V2X messages	PQC-only for infrastructure (CA, OTA, V2C backend)
Hybrid PQC/ECC with no latency degradation	Hybrid ECC + ML-KEM for session key exchange
PQC HSMs that match current ECC performance	V2V safety beaconing still on ECC through 2030+
Drop-in algorithm replacement in existing stacks	New vehicles (post-2028) ship with PQC-capable HSMs

Seamless cross-border PQC certificate interoperability	Fragmented regional PQC timelines for 5–10 years

The hybrid approach is not a compromise it is the correct design

Running ML-KEM alongside ECDH, or ML-DSA alongside ECDSA, during the transition period provides security against both classical and quantum adversaries simultaneously. If either algorithm is unexpectedly broken, the other provides fallback security. The IETF is already standardising hybrid key exchange for TLS 1.3. V2X standards bodies should adopt the same principle.

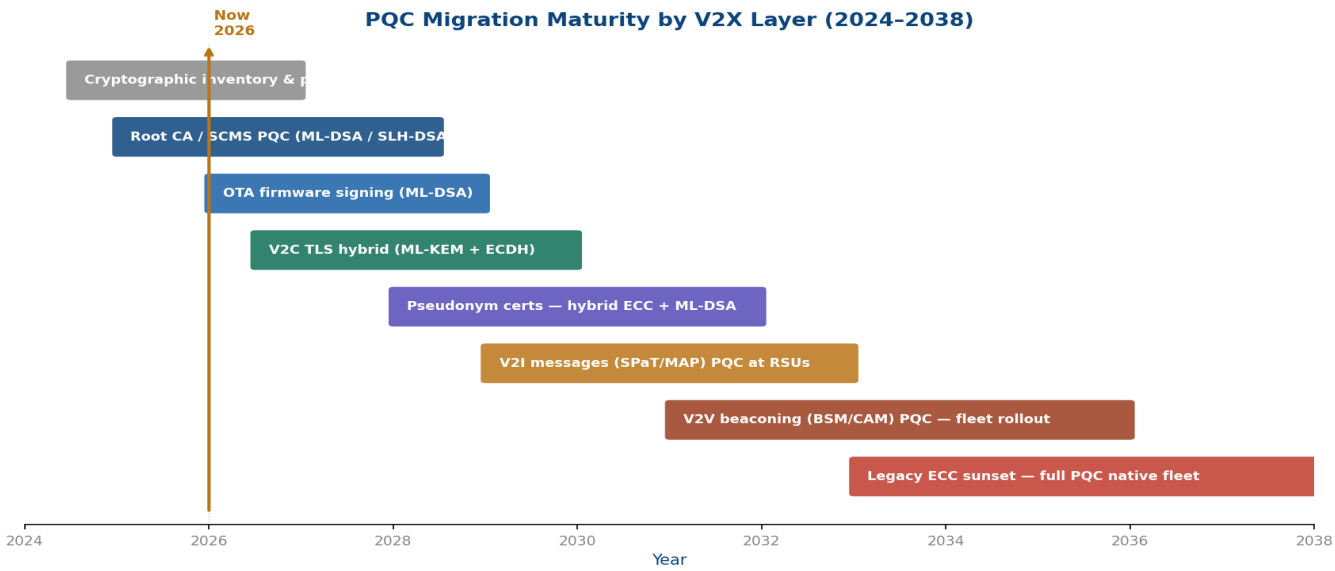

Figure 8.1 PQC migration maturity by V2X layer (2024–2038). Infrastructure layers migrate first; V2V beaconing trails significantly due to hardware and latency constraints.

09 Indicative migration roadmap

The following roadmap reflects a phased, infrastructure-first approach. Timelines are indicative and will vary by region and organisational readiness.

Table 7: PQC migration roadmap for V2X

Phase	Period	Key Actions
Phase 1- Inventory & Foundation	Now – 2026	Cryptographic asset inventory across all V2X components. Classify by sensitivity and lifecycle. Begin planning the SCMS/CCMS root CA PQC transition. Enable PQC hybrid in V2C TLS (ML-KEM + ECDH). Mandate PQC-capable HSMs in all new OBU RFPs.
Phase 2 - Infrastructure PQC	2026 – 2028	PQC root and sub-CA deployment in SCMS/CCMS. OTA update signing migrated to ML-DSA or SLH-DSA. Pilot PQC-capable RSUs in smart corridor zones. IEEE/ETSI standards working groups produce proposals for PQC-compatible certificate formats.
Phase 3 - Fleet Hybrid Rollout	2028 – 2031	Pseudonym certificate issuance has been migrated to a hybrid ECC + ML-DSA model. V2I messages (SPaT, MAP, DENM) transition to PQC-signed formats at upgraded RSUs. Dual-stack RSU verification required. Regional cross-border PQC trust agreements established.

Phase 4 - V2V Migration	2031 – 2035	V2V beaconing (BSM/CAM) transitions to PQC as fleet PQC-OBU share exceeds threshold. ECC sunset begins at the infrastructure level. Standards bodies publish PQC-native V2X security profiles. Legacy fleet maintained on ECC during run-out.
Phase 5 - Full PQC Native	Post-2035	ECC deprecated per NIST IR 8547 timeline. New vehicle platforms operate natively on PQC. Legacy hybrid-support maintained until fleet turnover completes. Crypto-agility provisions for future algorithm updates.

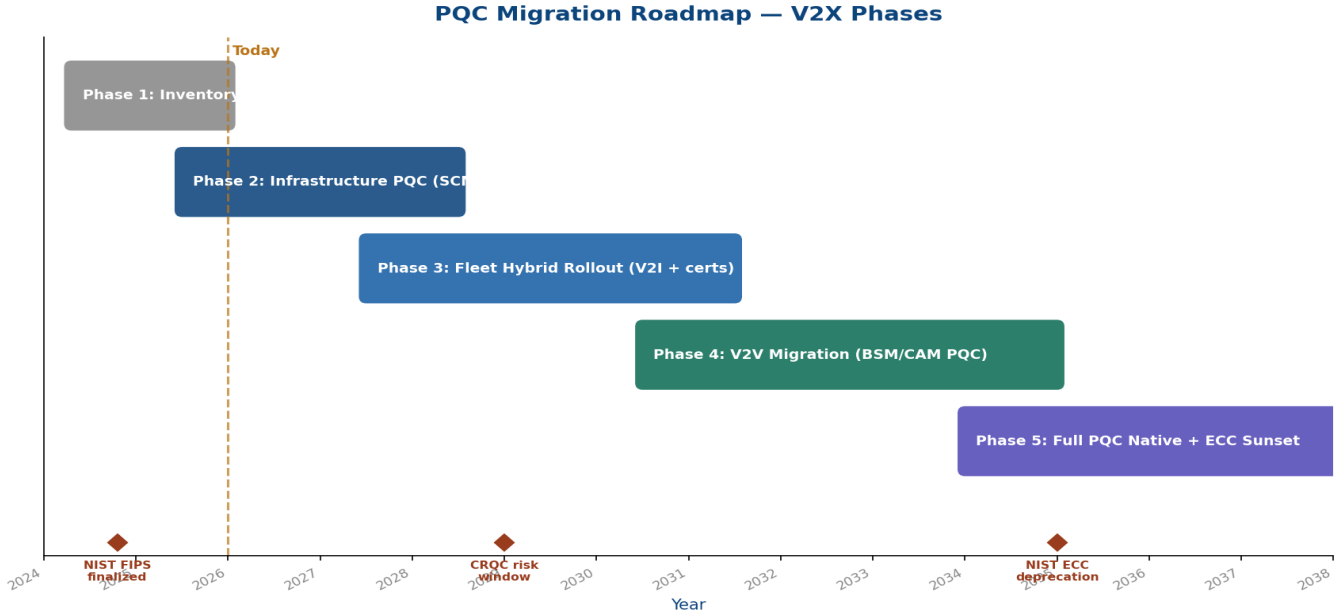

Figure 9.1 PQC migration roadmap: five phases from inventory (2024) through full PQC-native operation (post-2035), with key milestones and NIST ECC deprecation marker.

10 The hard questions - What challenges does the approach face

Any credible treatment of this topic must confront the questions that optimistic roadmaps tend to paper over. These are the open questions that remain structurally unresolved.

Is the V2V beaconing problem actually solvable within real-time constraints?

Current research shows that even optimised PQC implementations struggle to meet the 10 ms verification target for V2V safety messages on automotive-grade embedded hardware without dedicated acceleration. Hardware acceleration for ML-DSA and FN-DSA on automotive-grade silicon remains nascent. If PQC signatures cannot be verified at 10 Hz with sub-10 ms latency, the safety case for V2X is not improved by PQC adoption; it only changes in character. This is the central unresolved question.

Who coordinates the SCMS root CA transition at scale?

The SCMS in North America and the CCMS in Europe are multi-stakeholder systems with government, industry, and regulatory dimensions. A root CA key transition is not a technical event; it is a governance event. No single entity owns the transition. The absence of a clear governance authority for an SCMS PQC migration is a larger obstacle than the technical challenges.

Does geographic scoping create interoperability dead zones?

If the EU migrates to PQC-native certificates in 2029 and the US does not complete that transition until 2033, vehicles crossing borders may encounter certificate validation failures. The answer is hybrid certificates and cross-domain trust anchors, but the standards for these do not yet exist. Cross-jurisdiction PQC interoperability is an active standardisation gap.

Are the NIST PQC algorithms the right ones for V2X specifically?

NIST selected ML-KEM, ML-DSA, and SLH-DSA as general-purpose algorithms. V2X is not a general-purpose environment; it is latency-critical, bandwidth-constrained, and hardware-limited. There is a legitimate research question about whether V2X-specific lightweight PQC variants would be more appropriate than direct adoption of NIST FIPS. Standardised algorithms are safer from a compliance and interoperability perspective, but they may not be optimal for the V2X performance envelope.

Summary: The clear path forward

1. Start with the infrastructure layer, not the vehicle layer. SCMS/CCMS, OTA pipelines, and V2C backend TLS are the high-value, achievable first steps.
2. Adopt hybrid cryptography (PQC + ECC) as the transition architecture. It is not a compromise; it is the correct design for long-lifecycle systems.
3. Define geographic deployment boundaries based on existing managed ITS zones (smart corridors, C-ITS deployment areas). These provide a natural scope for pilot migration.
4. Mandate PQC-capable HSMs in all new OBU/RSU hardware procurements from 2026 onwards. The silicon must be capable, even if not yet activated.
5. Push for standards body action on PQC-compatible certificate formats in IEEE 1609.2 and ETSI TS 103 097. Certificate format redesign is the long-lead item.
6. Accept that V2V safety message authentication will remain on ECC or hybrid ECC/PQC through 2030+, design the architecture accordingly, with a planned sunset.
7. Treat crypto-agility as a first-class design requirement. The V2X security architecture must be updatable without requiring a full hardware refresh of the fleet.
8. Address the governance gap in SCMS/CCMS PQC migration early. Technical readiness without governance coordination achieves nothing at scale.

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Sumit Chouhan is a cybersecurity professional in leading enterprise, automotive, and critical infrastructure security programs across complex, highly regulated environments. Sumit writes on enterprise cybersecurity strategy, post-quantum readiness, V2X, EVSE security architecture, and the governance challenges. His analysis is intended for security practitioners, standards contributors, and technology leaders who need technically grounded, unvarnished perspectives on where the field is headed.

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References

All information used, gathered or otherwise brought forward in this article is listed below.

Standards & Regulatory

R01 **NIST Post-Quantum Cryptography Standardisation**
NIST / CSRC
Official NIST project page covering ML-KEM (FIPS 203), ML-DSA (FIPS 204), SLH-DSA (FIPS 205), and FN-DSA (FIPS 206). The authoritative source for all PQC algorithm specifications and migration guidance.
Link: https://csrc.nist.gov/projects/post-quantum-cryptography

R02 **NIST IR 8547: Transition to Post-Quantum Cryptography Standards**
NIST
Defines the transition timeline: deprecation of quantum-vulnerable algorithms by 2030 for high-risk systems, full deprecation by 2035. Mandatory reading for any V2X migration planning.
Link: https://csrc.nist.gov/pubs/ir/8547/ipd

R03 **FIPS 203 Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM)**
NIST
Full specification of ML-KEM (Kyber). The primary algorithm for post-quantum key establishment, replacing ECDH and ECIES in V2X session key exchange and V2C TLS.
Link: https://doi.org/10.6028/NIST.FIPS.203

R04 **FIPS 204 Module-Lattice-Based Digital Signature Standard (ML-DSA)**
NIST
Full specification of ML-DSA (Dilithium). Primary post-quantum digital signature algorithm for certificate signing, OTA firmware signing, and V2I message authentication.
Link: https://doi.org/10.6028/NIST.FIPS.204

R05 **FIPS 205 Stateless Hash-Based Digital Signature Standard (SLH-DSA)**
NIST
Full specification of SLH-DSA (SPHINCS+). Hash-based signature scheme best suited to root/sub-CA signing in V2X PKI. Signature sizes (8–50 KB) preclude use in high-frequency V2V messaging.
Link: https://doi.org/10.6028/NIST.FIPS.205

R06 **IEEE 1609.2-2022 Wireless Access in Vehicular Environments: Security Services**
IEEE
The foundational V2X security standard defining WAVE message formats, ECDSA-based certificate structures, and the pseudonym certificate model. The standard whose certificate format will need redesign to accommodate PQC.
Link: https://standards.ieee.org/ieee/1609.2/10323/

R07 **ETSI TS 103 097 Intelligent Transport Systems: Security Header and Certificate Formats**
ETSI
European equivalent to IEEE 1609.2, specifying C-ITS security architecture including ECDSA-based certificate formats for CAM, DENM, SPaT and other ITS messages.

Link: https://www.etsi.org/deliver/etsi_ts/103000_103099/103097/

R08

CISA Product Categories for Technologies Using PQC Standards

CISA / DHS

The US government's list of product categories where PQC-capable products are now widely available. Relevant to V2X hardware procurement agencies and operators, they should acquire only PQC-capable OBUs and RSUs going forward.

Link: https://www.cisa.gov/resources-tools/resources/product-categories-technologies-use-post-quantum-cryptography-standards

R09

ISO/SAE 21434:2021 Road Vehicles: Cybersecurity Engineering

ISO / SAE

Process-oriented automotive cybersecurity standard. A PQC migration constitutes a cybersecurity-relevant change under this framework and requires documented risk assessment and updates to the cybersecurity management system.

Link: https://www.iso.org/standard/70918.html

Technical research papers

R10

Post-Quantum Cryptography for Intelligent Transportation Systems

arXiv / Preprint (2025)

Comprehensive survey of how NIST PQC standards map onto current V2X security frameworks, including IEEE 1609.2, ETSI C-ITS, and 3GPP C-V2X. Includes a comparative overview table of standards and PQC readiness.

Link: https://d197for5662m48.cloudfront.net/documents/publicationstatus/291831/preprint_pdf/d8c11fbe5864c46da274949a3c4310e3.pdf

R11

AI-Driven Post-Quantum Cryptography for Cyber-Resilient V2X Communication

arXiv (2025)

Demonstrates that standard PQC solutions struggle to meet the 10 ms V2V latency requirement, and proposes AI-driven adaptive algorithm selection using reinforcement learning as a mitigation strategy.

Link: https://arxiv.org/html/2510.08496v1

R12

Adaptive Quantum-Safe Cryptography for 6G Vehicular Networks via Context-Aware Optimisation

arXiv (2026)

Proposes the APMOEA framework using reinforcement learning to adapt PQC algorithm selection dynamically under vehicular URLLC constraints. Reports 27% latency reduction over static PQC deployment.

Link: https://arxiv.org/html/2602.01342v1

R13

A Lightweight Key Agreement Protocol for V2X Based on Kyber and Saber

MDPI Sensors (2025)

Proposes a hybrid Kyber + Saber key agreement protocol for V2X. Validates on Veins + SUMO simulation; reports 60% handshake latency reduction vs RSA and ~40% lower resource consumption, a strong candidate for practical deployment.

Link: https://www.mdpi.com/1424-8220/25/22/6938

R14

Post-Quantum Cryptography Challenges in Connected Vehicles with V2X

ResearchGate (2024)

Evaluates PQC performance under high-density vehicular traffic conditions. Reports latency reduced by 18% and throughput increased by 22% using an FL + blockchain security framework vs traditional ECC methods.

Link: https://www.researchgate.net/publication/382709928_Post-Quantum_Cryptography_Challenges_in_Connected_Vehicles_with_V2X

R15

Machine Learning-Based Blockchain Technology for Secure V2X Communication

MDPI Sensors (2025)

Surveys integration of ML and blockchain for adaptive, decentralised V2X security. Identifies PQC's key size and signature size as the principal obstacles to direct adoption in V2V beaconing environments.

Link: https://www.mdpi.com/1424-8220/25/15/4793

R16

Comparative Experiments of V2X Security Protocol Based on Hash Chain Cryptography

PMC / NCBI

Empirical comparison of IEEE 1609.2 and ETSI-103-097 latency using commercial DSRC hardware (Cohda MK5). Reports ECDSA end-to-end delay of 10–65 ms the baseline against which PQC overhead must be measured.

Link: https://pmc.ncbi.nlm.nih.gov/articles/PMC7582867/

R17

Post-Quantum Cryptography for Automotive Systems

ScienceDirect / Computers & Electrical Engineering (2021)

Early and frequently cited paper examining lattice-based PQC for automotive embedded systems. Benchmarks NIST round-3 finalists on automotive-grade hardware, establishing the performance baseline for subsequent research.

Link: https://www.sciencedirect.com/science/article/abs/pii/S0141933121005299

R18

A Comprehensive Survey of V2X Cybersecurity Mechanisms

IEEE Open Journal of Communications Society / FIREMAN Project

Extensive taxonomy of V2X attack classes, security requirements, and countermeasures. Covers classical and emerging threats including quantum attacks. Recommended as a foundational reference for V2X security programme design.

Link: https://fireman-project.eu/attachments/article/25/_Camera_ready_final____IEEE_OJ_CommSoc__V2X_security_survey.pdf

R19

V2X Credential Management System Comparison: IEEE 1609.2.1 vs ETSI TS 102 941

IEEE Xplore (2024)

Compares the North American SCMS and European CCMS architectures in detail certificate application processes, PKI structures, and message formats. Essential context for understanding the scale of a PQC root CA migration.

Link: https://ieeexplore.ieee.org/document/10774595

R20

The Impact of ECDSA in a VANET Routing Service

ScienceDirect / Ad Hoc Networks (2019)

Measures ECDSA computational overhead in a real deployed VANET (city of Porto, Portugal). Demonstrates that ECDSA authentication at 10 Hz can disrupt routing in sparse networks directly motivating the need for lighter-weight PQC alternatives.

Link: https://www.sciencedirect.com/science/article/abs/pii/S1570870518306127

Industry Reference

R21

Quantum Threats Meet the Connected Car

TechArena (2026)

Industry-focused analysis of quantum risk across the connected vehicle attack surface WiFi, Bluetooth, TPMS, OBD-II, EV charging, and V2X. Frames PQC adoption within ISO 21434 compliance and the Software Defined Vehicle transition.

Link: https://techarena.ai/content/quantum-threats-meet-the-connected-car

R22

A Complete Guide to Post-Quantum Cryptography Standards

Palo Alto Networks

Accessible industry reference covering PQC standards by region, algorithm selection guidance, and migration planning. Highlights divergence between NIST, ETSI, and national standards bodies directly relevant to the cross-border interoperability challenge.

Link: https://www.paloaltonetworks.com/cyberpedia/pqc-standards

R23

Post-Quantum Cryptography (PQC) Compliance Standards

SafeLogic

Reference guide for US federal PQC compliance timelines under OMB, CISA, and NSA guidance. Covers CNSA 1.0, CNSA 2.0, and the Quantum Cybersecurity Preparedness Act, the regulatory driver for V2X infrastructure operators.

Link: https://www.safelogic.com/compliance/pqc-standards

R24

Quantum-Resistant Cryptography for Automotive Cybersecurity

Cybersecurity and Network Defense Research (2023)

Proposes a layered PQC architecture for V2X combining lightweight PQC with classical methods to ensure forward secrecy and smooth transition without requiring full protocol redesign. Includes a simulation case study of lattice-based key exchange in an AV network.

Link: https://thesciencebrigade.com/cndr/article/view/364

R25

Ensuring Security and Privacy in VANET: Authentication Approaches Survey

Wiley / Journal of Computer Networks and Communications (2024)

Comprehensive 2024 survey of VANET authentication schemes. Covers ECDSA, symmetric key, identity-based, and emerging PQC approaches. Useful reference for understanding the full authentication landscape before committing to a PQC migration strategy.

Link: https://onlinelibrary.wiley.com/doi/full/10.1155/2024/1818079

Stay current

The PQC landscape is moving fast. NIST continues to publish updates at csrc.nist.gov/projects/post-quantum-cryptography. ETSI TC ITS WG5 and the IEEE 1609.2 working group are the primary venues for V2X-specific PQC standardisation progress. For automotive cybersecurity governance, ISO/SAE 21434 and UNECE WP.29 R155 remain the regulatory anchors.